

OVER-SPEED ALARM FOR TWO-WHEELERS

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Two-wheeler riding calls for constant attention. The moment you become unmindful of the high speed, applying sudden brakes to avoid obstacles may lead to an accident.

An over-speed alarm can be of great help to avoid such situations. It acts like a watchful friend that warns you whenever you drive faster than the set speed limit. Here we present the circuit of a simple and easy-to-use

PARTS LIST

Semiconductors:

IC1	- A1302 Hall sensor
IC2	- LM358 operational amplifier
IC3	- LM2917N frequency-to-voltage converter
IC4	- TL071 low-noise operational amplifier
IC5	- CA3130 operational amplifier
IC6	- UM66 music generator
IC7	- 7805, 5V regulator
T1	- BC548 npn transistor
D1	- 1N4148 switching diode
D2	- 1N4007 rectifier diode
ZD1	- 3.3V zener diode
LED1, LED2	- 5mm LED

Resistors (all 1/4-watt, 5% carbon)

R1, R13	- 220-ohm
R2	- 4.7-kilo-ohm
R3, R4	- 470-ohm
R5	- 22-kilo-ohm
R6-R8	- 10-kilo-ohm
R9, R10	- 560-kilo-ohm
R11	- 100-ohm
R12	- 330-ohm
R14	- 470-ohm
VR1	- 2-kilo-ohm
VR2	- 100-kilo-ohm
VR3	- 1-kilo-ohm
VR4	- 10-kilo-ohm

Capacitors:

C1	- 2.2µF, 16V electrolytic
C2	- 22nF ceramic disk
C3	- 0.47µF, 35V electrolytic
C4	- 220µF, 35V electrolytic
C5, C8	- 0.1µF ceramic disk
C6, C7	- 470µF, 35V electrolytic
C9	- 220µF, 16V electrolytic

Miscellaneous:

LS1	- 8-ohm, 1W loudspeaker
BATT.	- 12V battery
	- Magnet

over-speed indicator based on Hall effect sensor.

Hall Effect

When a current-carrying conductor is placed in a magnetic field, a voltage is generated perpendicular to both the current and the field. This is known as the Hall effect.

Fig. 1 shows the basic principle of the Hall effect. Current is passed through a thin sheet of semiconducting

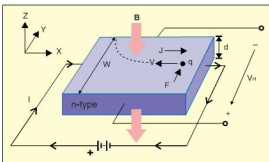


Fig. 1: Hall effect principle

material (Hall element) in a direction perpendicular to the output connections. When no magnetic field is present, current distribution is uniform with no potential difference developed across the output. When a perpendicular magnetic field is applied, a Lorentz force (force acting on the moving charge carriers in a magnetic field) disturbs the current distribution, resulting in a potential difference (voltage) across the output. This voltage is called the Hall voltage (V_H).

Circuit description

Fig. 2 shows the circuit of the over-speed alarm for two-wheelers. It consists of a Hall sensor, operational amplifiers, frequency-to-voltage converter, music generator and regulator.

The A1302 used here is a continuous-time, ratiometric, linear Hall effect

sensor IC. It is optimised to accurately provide a voltage output that is proportional to an applied magnetic field. The Hall effect integrated circuit includes a Hall circuit, a linear amplifier and a CMOS Class A output structure. Integrating the Hall circuit and the amplifier on a single chip minimises many of the problems normally associated with low-voltage-level analogue signals. High precision in output levels is obtained by internal gain and offset trim adjustments made at the end-of-line during the manufacturing process.

Hall sensor A1302 (IC1) gives an output proportional to the magnetic field intensity, which is converted into pulse by op-amp LM358. LED1 glows to indicate a high pulse output. The sensitivity of op-amp LM358 can be adjusted using preset VR1.

Applying a south-polarity (+B) magnetic field perpendicular to the branded face (front view) of the sensor A1302 increases its output voltage. Conversely, applying a north-polarity (-B) magnetic field perpendicular to the branded face decreases the sensor output.

The output of LM358 (IC2) is applied to frequency-to-voltage converter LM2917N (IC3). LM2917N consists of an input amplifier with built-in hysteresis, a charge pump frequency-to-voltage converter, a versatile op-amp/comparator with an uncommitted output transistor, and an active zener regulator. The inbuilt charge pump technique of LM2917N converts the frequency into a proportional voltage with the help of external passive components (capacitor C3 and preset VR2), which is available at shorted junction of pins 5 and 10. Preset VR2 is used to

